

CS & IT ENGINEERING



Computer Network

Principles of Layering

Lecture No. - 12



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Recap of Previous Lecture



Topic

End-to-End Delays





Topics to be Covered



Topic

Principles of Layering

* Concepts of Layering





Topic : Computer Network



→ Define communication of hosts

→ Host : Computer, Server or Smartphone etc.



Topic : Network Communication Model

→ Two famous model :-

1. TCP/IP Model → DARPA
2. OSI Model → ISO

OSI Model is superset



Topic : TCP/IP Model



- Internet protocol stack →
- Conceptual Model
- Layered Architecture

Layer
→ group of $f(n)$ s



Topic : TCP/IP Model



→ 4 - Layer Model

Application Layer

Transport Layer

Internet Layer

Network Access Layer



Topic : TCP/IP Model



→ 5 - Layer Model

Application Layer

Transport Layer

Internet Layer

Link Layer

Physical Layer



Topic : OSI Model



- ISO-OSI Model
- Open System Interconnection
- Conceptual Model



Topic : OSI Model

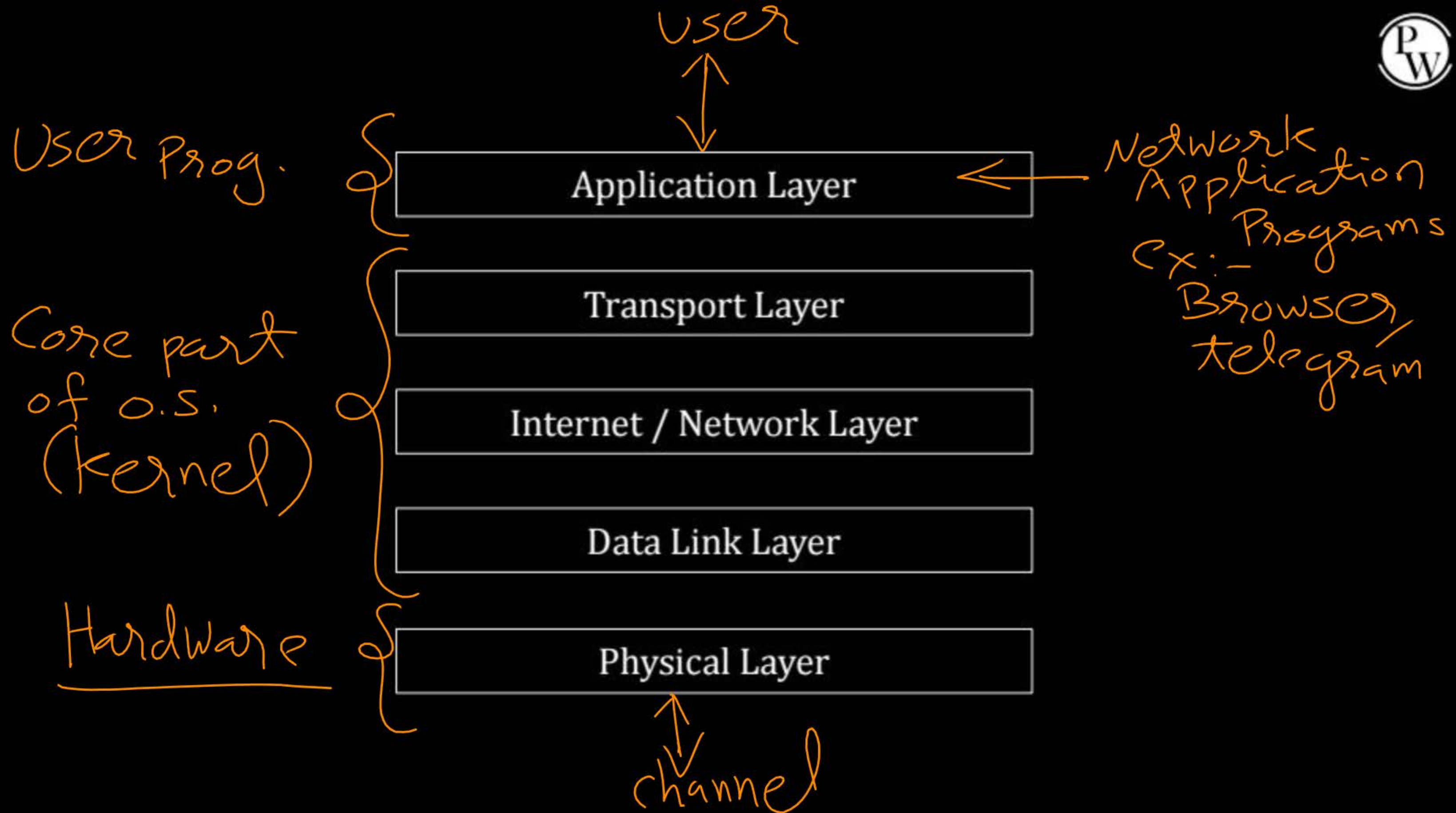


→ 7 - Layer Model

Host Layer

Layer 7	Application Layer
Layer 6	Presentation Layer
Layer 5	Session Layer
Layer 4	Transport Layer
Layer 3	Network Layer
Layer 2	Data Link Layer
Layer 1	Physical Layer

Media Layer





Topic : Transport Layer

V. V. Imp.



Application Layer Protocol	Transport Layer Protocol
<u>DNS</u>	<u>UDP</u> (default), <u>TCP</u> (conditional)
✓ <u>HTTP 1 & HTTP 2</u>	→ TCP
* HTTP 3 → QUIC	→ UDP
<u>FTP, SMTP, IMAP (POP)</u>	→ TCP
<u>DHCP, SNMP</u>	→ UDP
✓ RIP	→ UDP
* BGP	→ TCP
Real-time Multimedia	→ UDP (default)

[MCQ] [GATE-2013] [1-Mark] IIT-B

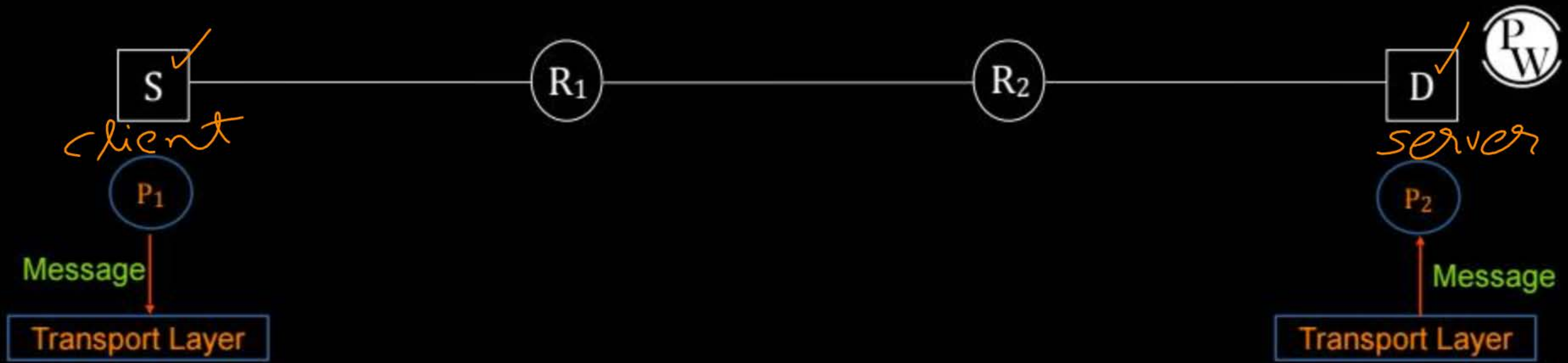


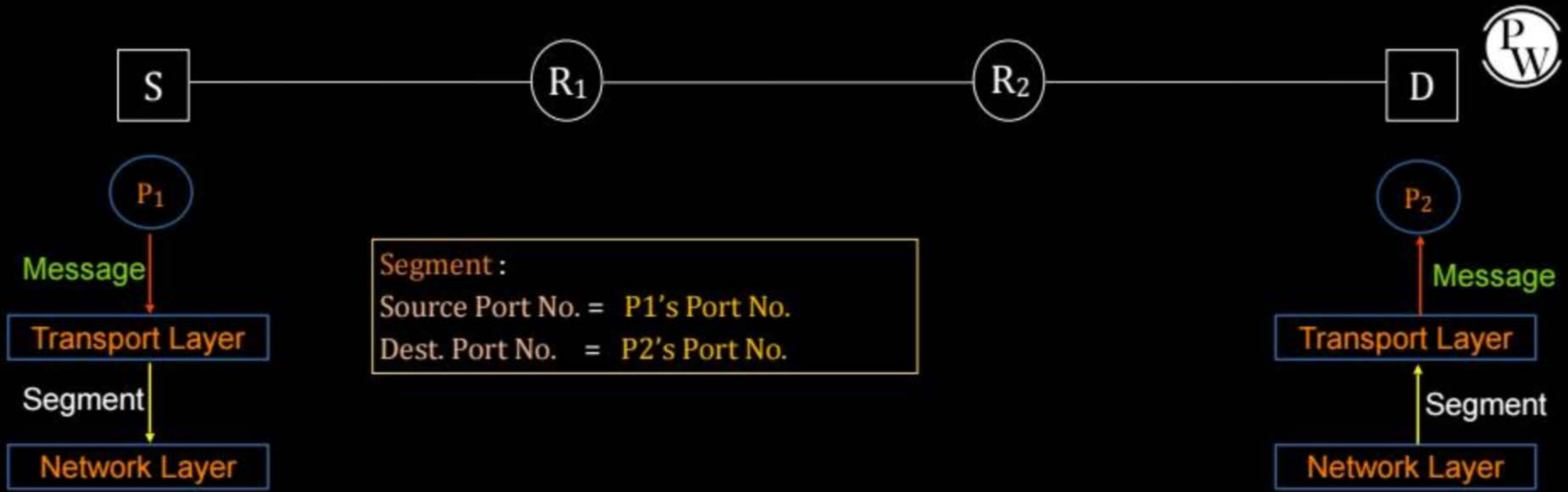
#Q. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D?

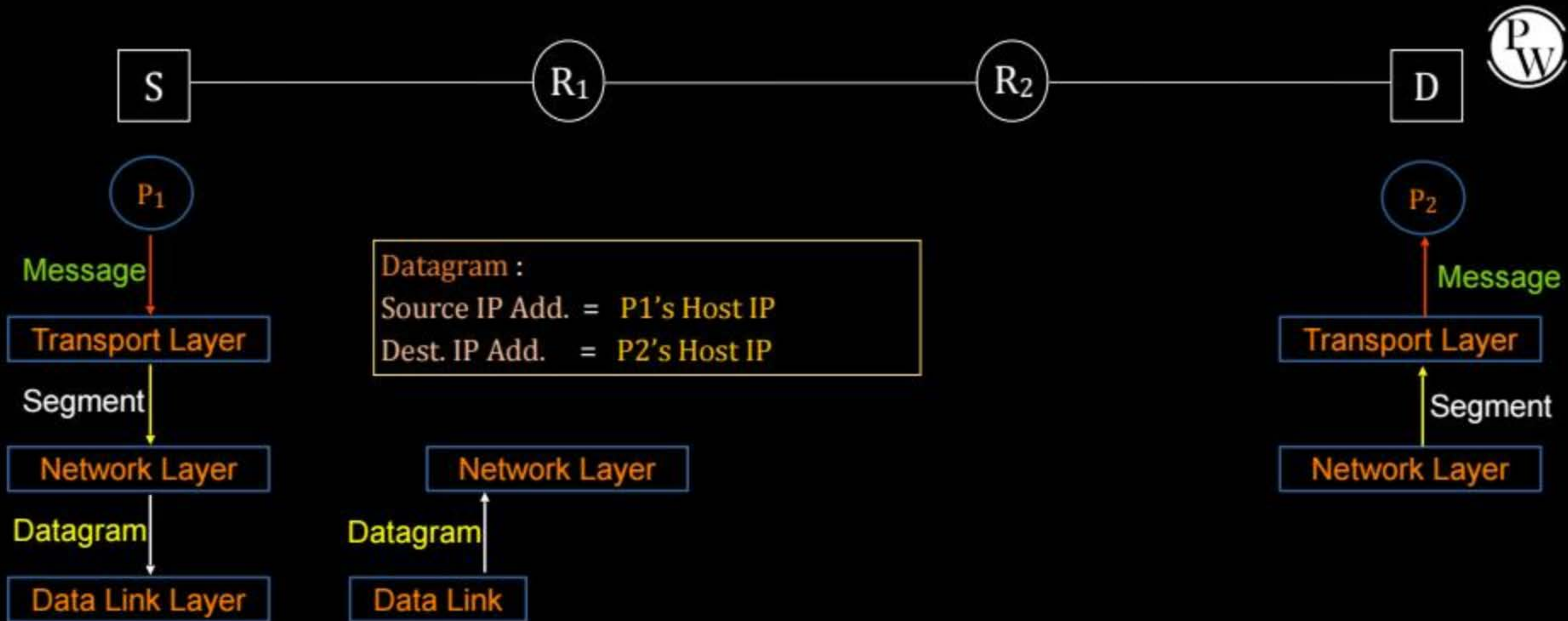


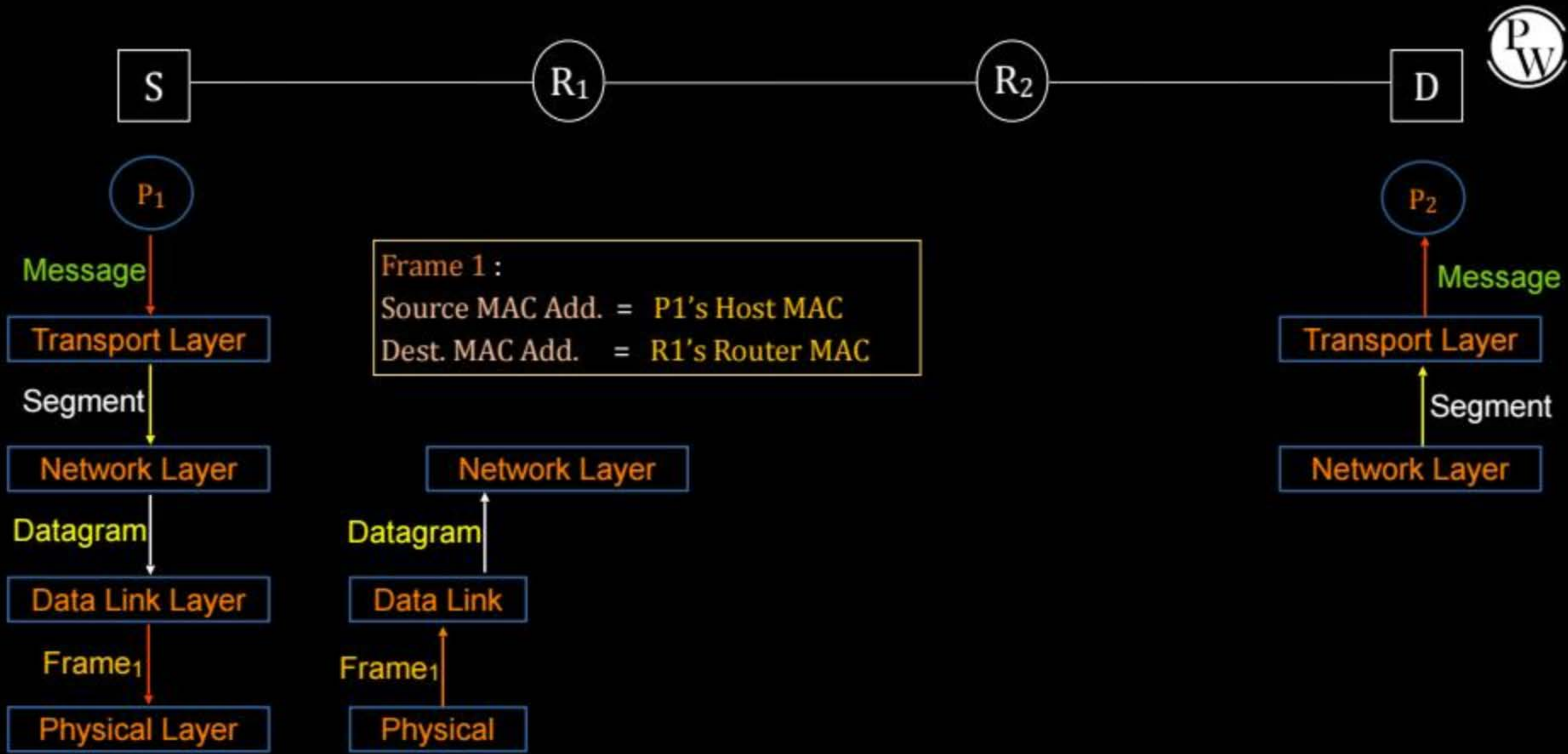
- A Network layer – 4 times and Data link layer – 3 times
- B Network layer – 4 times and Data link layer – 4 times
- ☒ C Network layer – 4 times and Data link layer – 6 times
- D Network layer – 2 times and Data link layer – 6 times

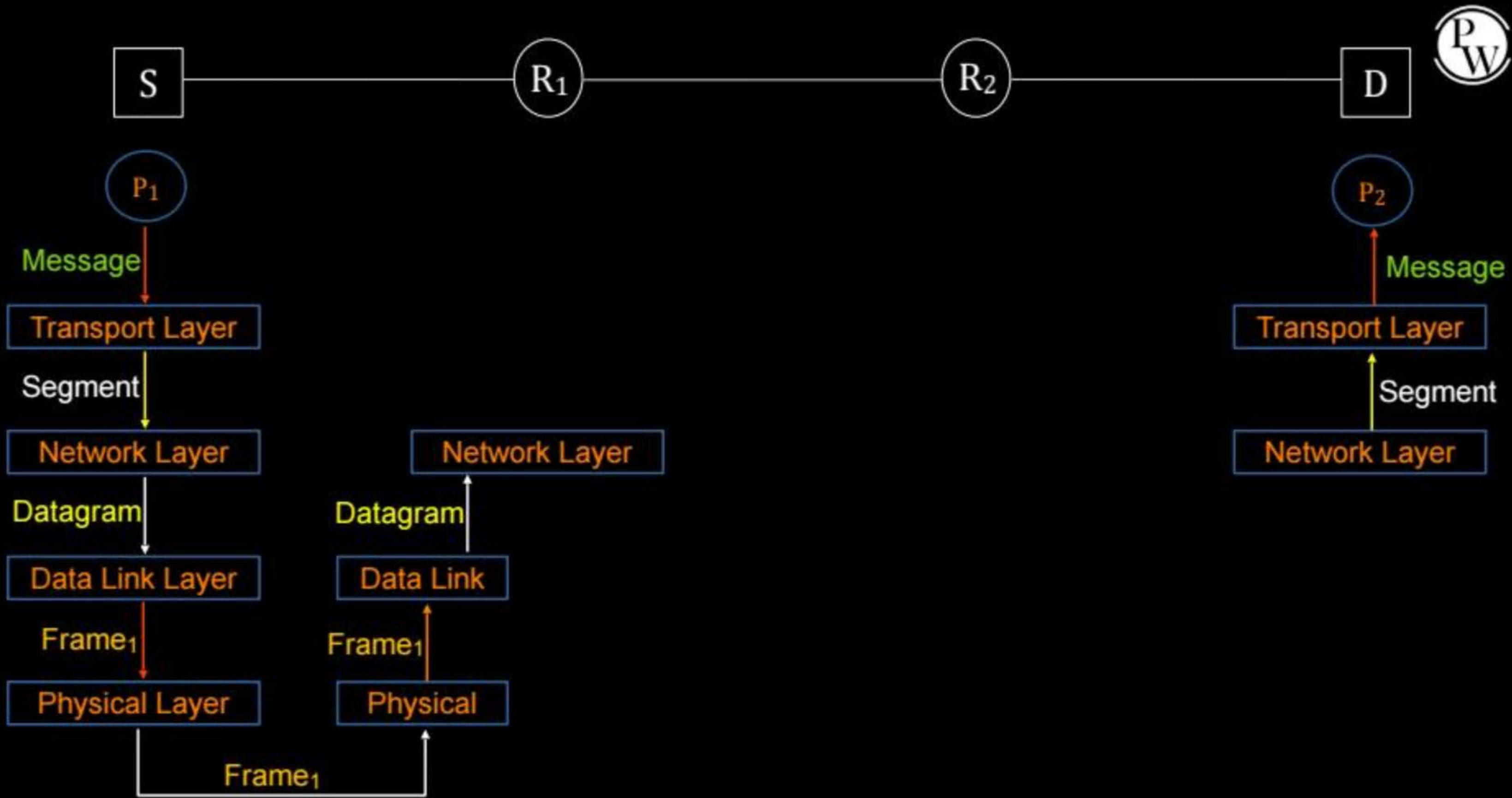
Ans: C

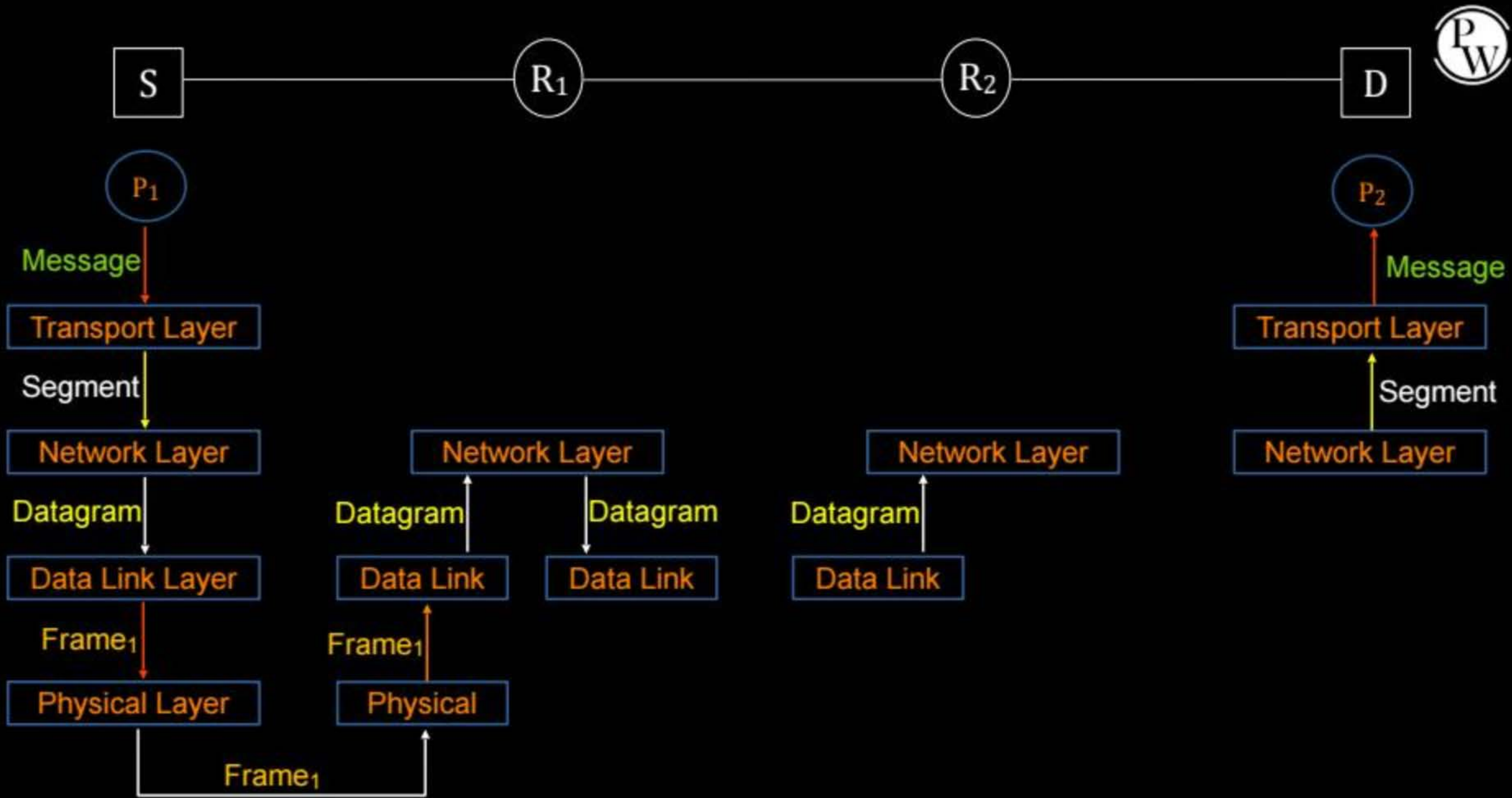


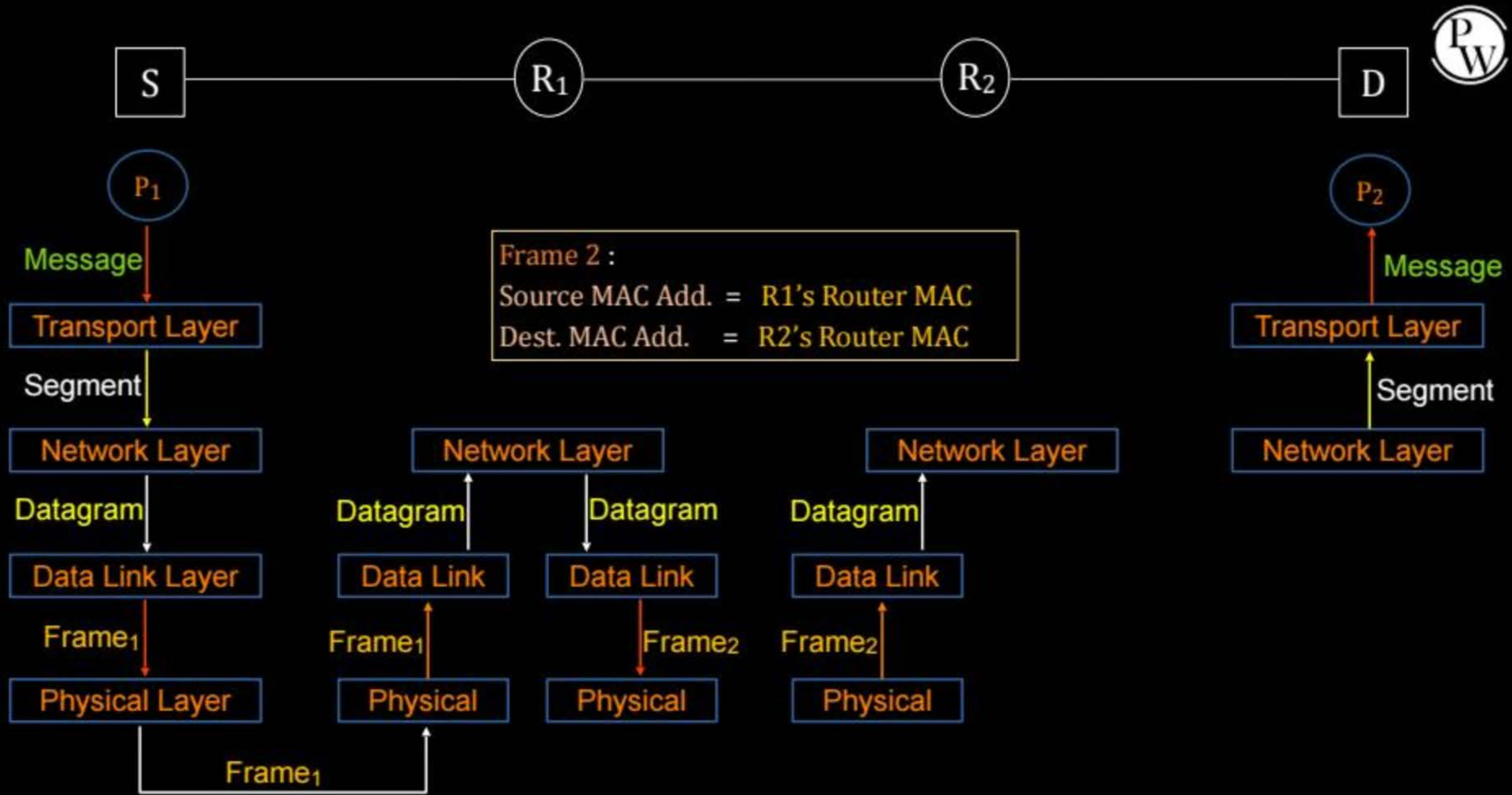


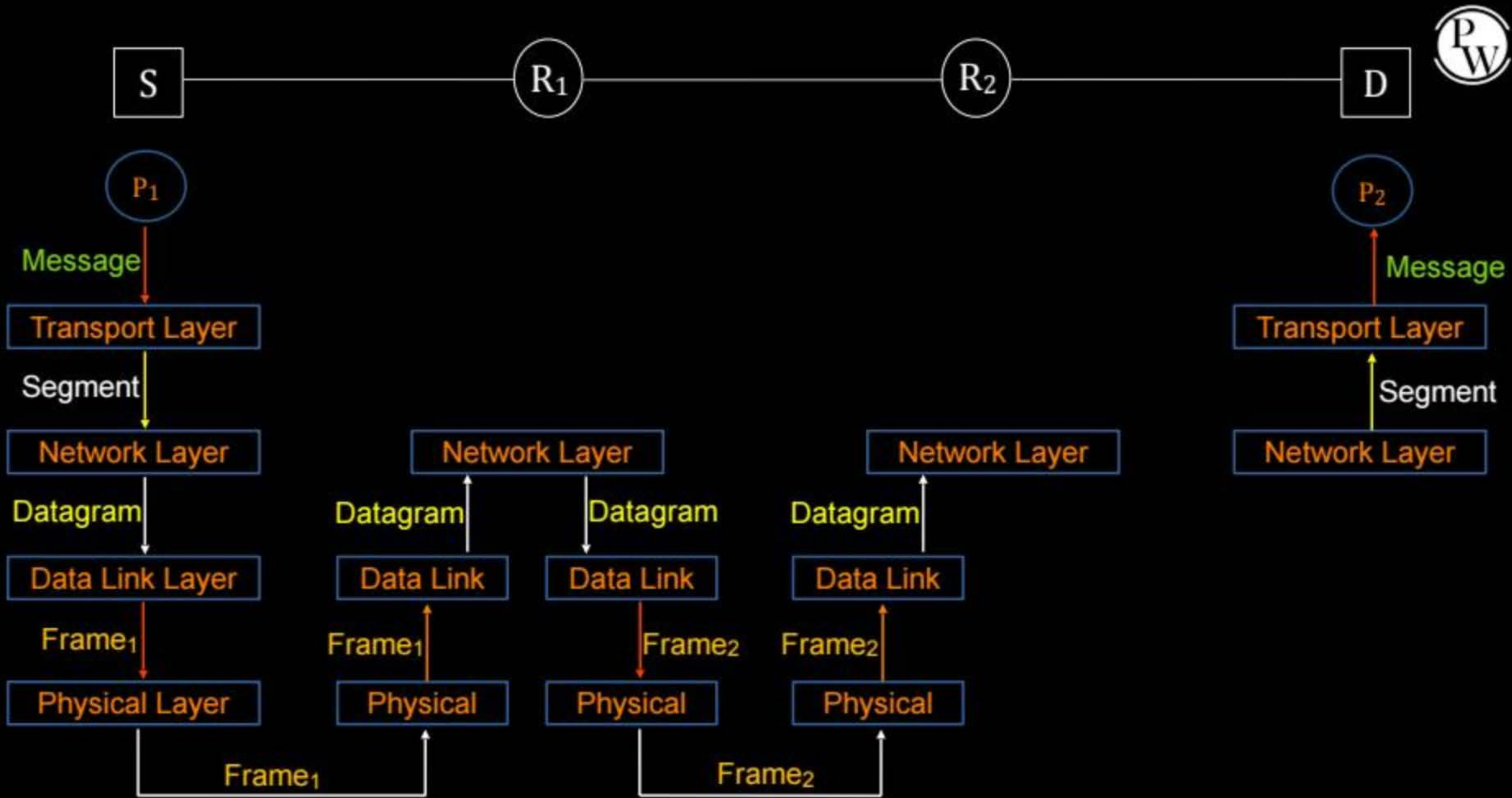


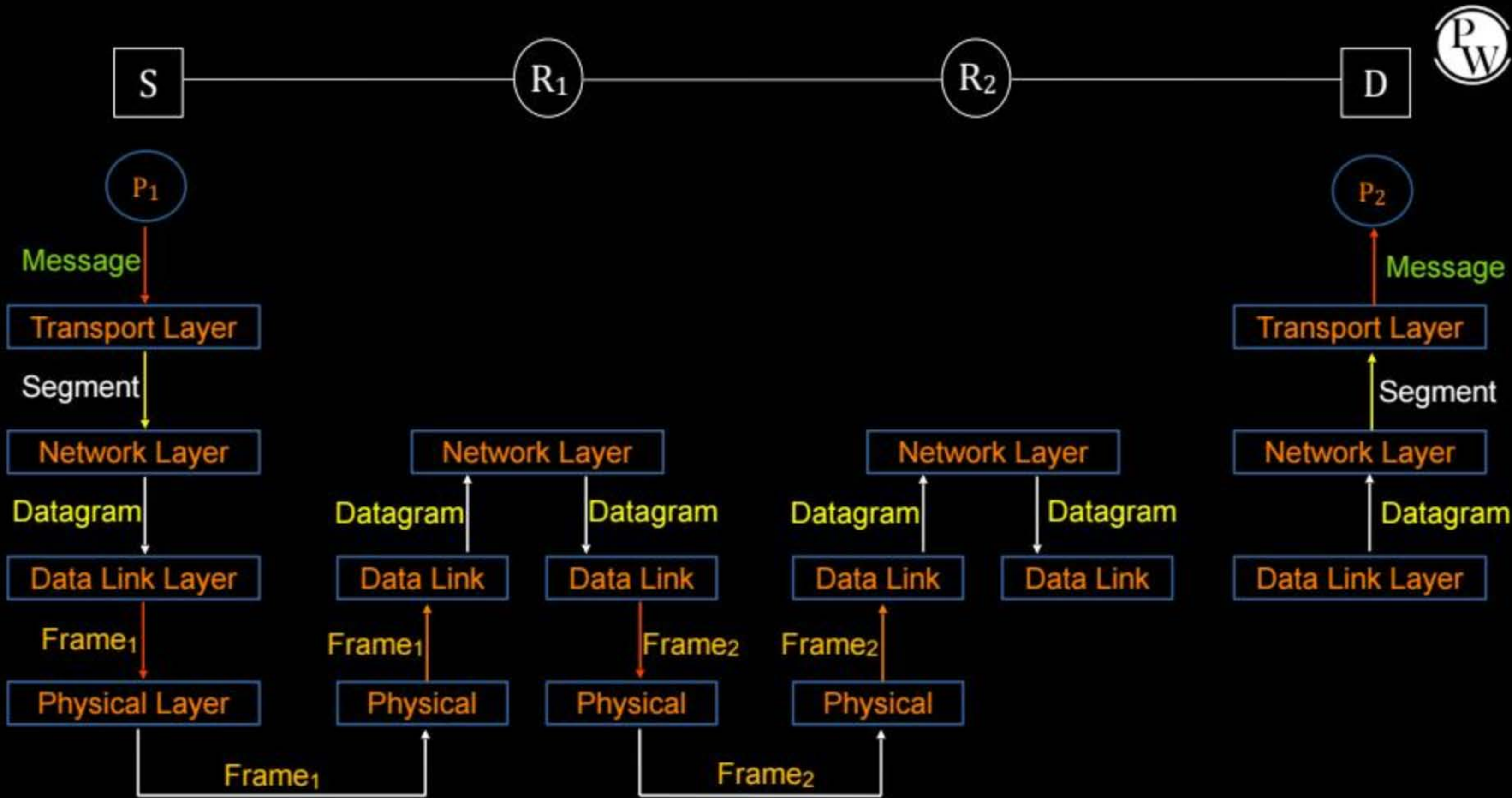


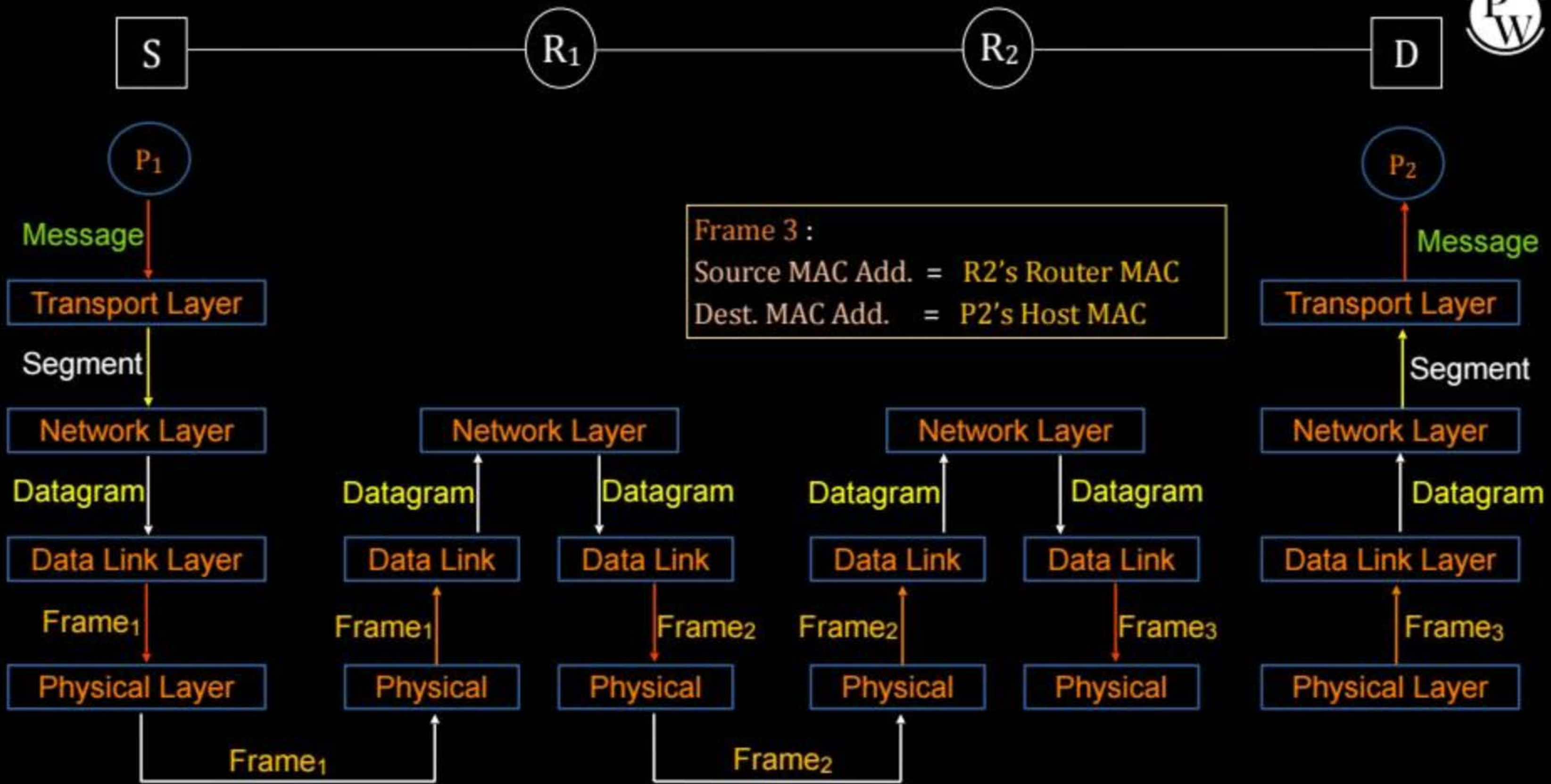


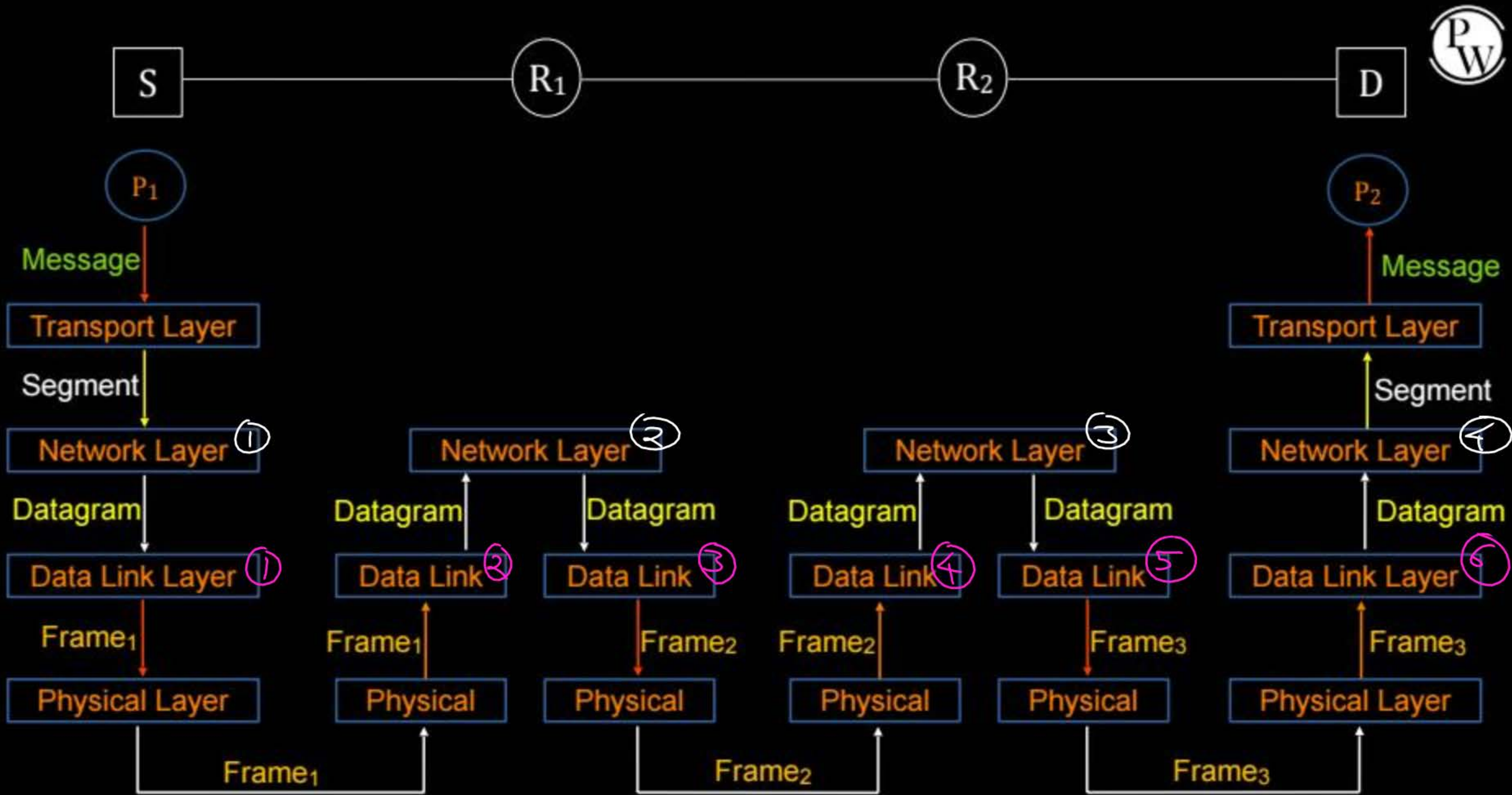












[MCQ] [GATE-2012] [1-Mark] IIT-D



#Q. The protocol data unit (PDU) for the application layer in the Internet stack is:

- ☐ A Frame
- ☐ B Datagram
- ☐ C Segment
- ☒ D Message

Ans: D



Topic : Protocol Data Unit

Layer	PDU
Application	Message
Transport	Segment
Network	Datagram
Data Link	Frame
Physical	Bit

(IP Pkt
IP fragment)



Topic : Layer Services



Layer	Services (PDU Creation)
Transport	Segmentation
Network	Fragmentation
Data Link	Framing



Topic : Address Type

Layer	Uses Type of Address
Transport	Port Number (<u>16-bits</u>)
Network	IPv4 Address (<u>32-bits</u>) or IPv6 Address (<u>128-bits</u>)
<u>Data Link</u>	MAC Address (<u>48-bits</u>)

logical
Address

physical
(Hardware)
Address



Topic : Layer Services



Layer	Provide Services (to its upper layer)
Transport	→ <u>Process-to-Process</u> [end-to-end] Connectivity
Network	→ <u>Host-to-Host</u> Connectivity (* Routing)
Data Link	→ <u>Node-to-Node</u> Connectivity

Routing ⇒ optimal path selection from source host to dest. host



Topic : Networking Device

Layer	Networking Device
Application	Gateway
Network	Router *
Data Link	<u>Switch</u> or <u>Bridge</u>
Physical	<u>Hub</u> , <u>Repeater</u>

→ Protocol Converter



Topic : Syllabus



→ Principles of Layering;

→ concept of layering
OSI & TCP/IP Model



Topic : Syllabus



→ Basics of switching (circuit, packet and virtual circuit) and performance metrics;

MUX
&
DeMUX
TDM/FDM

Network Delay
* End-to-End Delay
* Store-and-Forward Delay
* Bottleneck Bandwidth



Topic : Syllabus



Random Access
↑ [ALOHA/CSMA]

CSMA/CD
V.V.IMP

→ Data link layer : error detection, Medium Access Control, Ethernet;

→ CRC

V.V.IMP

* Hamming Distance

checksum, 2D parity, Hamming Code
(2008) (2021)

* Framing → Frame Synchronization
* Bit stuffing

* Ethernet bridging



Topic : Syllabus



→ Distance vector and link state routing;

Dynamic Routing

* Static Routing

Flooding, shortest path Algo.



Topic : Syllabus



→ IPv4 - Fragmentation, (V.V.Imp.)

↳ IPv4 Header



Topic : Syllabus



→ NAT → CS-24

→ CIDR Notation, Network Address Translation;

→ classless IPv4

* classful IPv4 → till 2008

* DHCP, ICMP, ARP



Topic : Syllabus



V.V.Imp.

→ TCP - flow control and congestion control, socket API;

*UDP



Topic : Syllabus



→ DNS and HTTP.
→ V.V. Imp.

* Email : SMTP
* FTP



2 mins Summary



Topic

Principles of Layering



THANK - YOU